

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Developed and implemented formal verification environments using **System Verilog Assertions (SVA)** for complex digital design blocks, achieving **95%** property coverage.
- Designed and integrated **transactional models** and **bus functional models (BFMs)** within verification testbenches to simulate NPU AI accelerator component interactions.
- Authored comprehensive test plans and executed diverse test cases, focusing on functional and formal verification methodologies for ASIC designs.
- Automated data pipeline tasks using **Python scripting**, processing large datasets for analysis and enhancing reporting efficiency by **30%**.
- Managed production support for enterprise applications, responding to P1/P2 incidents and ensuring SLA adherence through proactive monitoring with **Splunk** and **Dynatrace**.
- Collaborated with cross-functional teams to gather requirements and support User Acceptance Testing (UAT) for system configuration updates, facilitating smooth change management.

SKILLS

Verification Languages – System Verilog, System Verilog Assertions (SVA)

Verification Methodologies – Formal Verification, Assertion-Based Verification, UVM

Programming & Scripting – Python, C++, Shell Scripting, TCL

Hardware Description Languages – System Verilog

Tools & Technologies – ASIC, SoC, FPGA, AMBA, NoC, Jenkins, Git, Jira, Docker, AWS

Data Analysis & BI – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

AI Accelerator Pre-Silicon Verification Suite

- Developed a formal verification framework using **System Verilog Assertions (SVA)** to validate critical logic blocks within an AI accelerator design.
- Implemented **high-level transactional models** and **checkers** to simulate data flow and ensure functional correctness across IP components.
- Generated detailed test plans and coverage reports, identifying and documenting design anomalies prior to silicon fabrication.

Automated Transaction Monitoring Dashboard

- Built a Python-based data processing pipeline to ingest and transform financial transaction data, integrating with **PostgreSQL**.
- Designed and deployed interactive dashboards in **Power BI**, visualizing key operational metrics and enabling real-time transaction monitoring for enterprise clients.
- Implemented automated alerting mechanisms for failed payment messages, reducing manual triage time by **40%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)